

weeks-1-and-2\equations-inequalities.py

```
1 '''
2 equation written in factored form
3 '''
4 (w - 9)(w + 1) = 0
5
6 #set each factor equal to zero:
7 w - 9 = 0
8 + 9 + 9
9 #-----
10 w = 9
11
12 w + 1 = 0
13 - 1 - 1
14 #-----
15 w = -1
16
17 '''
18 equation written in factored form
19 '''
20 (9 - y)(4y - 2) = 0
21
22 #set each factor equal to zero:
23 9 - y = 0
24 - 9 - 9
25 #-----
26 [y = 9]
27
28 4y - 2 = 0
29 + 2 + 2
30 #-----
31 4y = 2
32 ÷ 4 ÷ 4
33 #-----
34 [y = (1 / 2)]
35
36 '''
37 solving equations in factored form
38 '''
39 (-5)(z)((z) + (-8))((z) + (-5)) = 0
40
41 #the factors are: -5, z, (z + (-8)), (z + (-5))
42 #set each factor equal to zero:
43 -5 = 0
44 #-----
45 #nope, -5 ≠ 0
46
47 z = 0
48 #-----
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49 [z = 0]
50
51 z + (-8) = 0
52 + 8 + 8
53 #-----
54 [z = 8]
55
56 z + (-5) = 0
57 + 5 + 5
58 #-----
59 [z = 5]
60
61 #final answer:
62 [z = 0, 8, 5]
63
64 '''
65 solving absolute value equations
66 '''
67 |w| - 19 = (-12)
68 + 19 + 19
69 #-----
70
71 [|w| = 7 and (-7)]
72
73 '''
74 next
75 '''
76 |w| + 11 = 7
77 - 11 - 11
78 #-----
79
80 [|w| = -4]
81
82 #nope, absolute value can never be negative
83 [no solution]
84
85 '''
86 next
87 '''
88 (-5)|y| = (-15)
89 ÷ (-5) ÷ (-5)
90 #-----
91
92 [|y| = 3 and (-3)]
93
94
95 '''
96 next
97 '''
98 (-4)|w| = 0
```

```

99  ÷ (-4)    ÷ (-4)
100 #-----
101
102 [|w|  =  0]
103
104
105 '''
106 next
107 '''
108      (-2)(y + 2) + 7  =  3(y + 6)
109      (-2y) + (-4) + 7  =  3y + 18
110
111      (-2y) + 3  =  3y + 18
112      + 2y          + 2y
113 #-----
114
115      3  =  5y + 18
116      - 18      - 18
117 #-----
118
119      (-15)  =  5y
120      / 5      / 5
121 #-----
122
123      [(-3)  =  y]
124
125 '''
126 next
127 '''
128      4(v + 1) + (-7)  =  3(v - 1) + v
129      4v + 4 + (-7)  =  3v + (-3) + v
130
131      4v + (-3)  =  4v + (-3)
132      - 4v          - 4v
133 #-----
134
135      [(-3)  =  (-3)]
136      [all real numbers are solutions]
137
138 '''
139 next
140 '''
141      (-3)(y + 1) + 7y  =  4(y + 1) + (-7)
142      (-3y) + (-3) + 7y  =  4y + 4 + (-7)
143
144      4y + (-3)  =  4y + (-3)
145      - 4y          - 4y
146 #-----
147
148      [(-3)  =  (-3)]

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149      [all real numbers are solutions]
150
151 '''
152 next
153 '''
154      2(v + 1) + 4v  =  3(2v - 1) + 8
155      2v + 2 + 4v  =  6v + (-3) + 8
156
157      6v + 2  =  6v + 5
158      - 6v          - 6v
159 #-----
160
161      [2  =  5]
162      [false statement; no solution]
163
164 '''
165 next
166 '''
167      5(2 + x) + (-x)  =  10 + 4(x + 1)
168      10 + 5x + (-x)  =  10 + 4x + 4
169
170      4x + 10  =  4x + 14
171      - 4x          - 4x
172 #-----
173
174      [10  =  14]
175      [false statement; no solution]
176
177 '''
178 next
179 '''
180      (-3)(w + 5) + 2  =  4(w + 2)
181      (-3w) + (-15) + 2  =  4w + 8
182
183      (-3w) + (-13)  =  4w + 8
184      + 3w          + 3w
185 #-----
186
187      (-13)  =  7w + 8
188      - 8          - 8
189 #-----
190
191      (-21)  =  7w
192      / 7      / 7
193 #-----
194
195      [(-3)  =  w]
196
197 '''
198 next

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199 '''
200     |2v + (-10)| = 4
201
202
203     2v + (-10) = 4
204     + 10      + 10
205 #-----
206     2v = 14
207     / 2    / 2
208 #-----
209     [v = 7]
210
211
212     2v + (-10) = (-4)
213     + 10      + 10
214 #-----
215     2v = 6
216     / 2    / 2
217 #-----
218     [v = 3]
219
220     [v = 7 and 3]
221
222 '''
223 next
224 '''
225     |3y + 6| = (-3)
226     [no solution]
227
228 '''
229 next
230 '''
231     |4x + 8| = 0
232
233     4x + 8 = 0
234     - 8   - 8
235 #-----
236
237     4x = (-8)
238     / 4    / 4
239 #-----
240
241     [x = (-2)]
242     [only one answer because 0 is neither positive nor negative]
243
244 '''
245 next
246 '''
247     factor →
248     x2 + 10x + 25

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249 '''
250 a2 + 2(a)(b) + b2 = (a + b)2
251 '''
252     x = (a)
253     5 = (b)
254
255     a2 + 2(a)(b) + b2 = (a + b)2
256     [x2 + 2(x)(5) + 52 = (x + 5)2]
257
258 '''
259 next
260 '''
261 '''
262 a2 + (-2)(a)(b) + b2 = (a - b)2
263 '''
264     u2 + (-8u) + 16
265
266     u = (a)
267     4 = (b)
268
269     a2 + (-2)(a)(b) + b2 = (a - b)2
270     [u2 + (-2)(u)(4) + 42 = (u - 4)2]
271
272 '''
273 next
274 '''
275 '''
276 a2 + (-2)(a)(b) + b2 = (a - b)2
277 '''
278     u2 + (-16u) + 64
279
280     u = (a)
281     8 = (b)
282
283     a2 + (-2)(a)(b) + b2 = (a - b)2
284     [u2 + (-2)(u)(8) + 82 = (u - 8)2]
285
286 '''
287 next
288 '''
289     x2 + 18x + 81
290
291     x = (a)
292     9 = (b)
293
294     a2 + 2(a)(b) + b2 = (a + b)2
295     [x2 + 2(x)(9) + 92 = (x + 9)2]
296
297 '''
298 next

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299 '''
300      $w^2 + 10w + 21 = 0$ 
301
302     #need two numbers that multiply to  $\rightarrow 21$  and add up to  $\rightarrow 10$ 
303      $a + b = 10$ 
304      $a * b = 21$ 
305
306      $(w + 3)(w + 7) = 0$ 
307      $w + 3 = 0$  or  $w + 7 = 0$ 
308      $w = (-3)$  or  $w = (-7)$ 
309      $[w = -3 \text{ and } -7]$ 
310
311 '''
312 next
313 '''
314      $y^2 + (-2)y + 1 = 0$ 
315
316     #need two numbers that multiply to  $\rightarrow 1$  and add up to  $\rightarrow (-2)$ 
317      $a + b = -2$ 
318      $a * b = 1$ 
319
320      $(y + (-1))(y + (-1)) = 0$ 
321      $y + (-1) = 0$  or  $y + (-1) = 0$ 
322      $y = 1$  or  $y = 1$ 
323      $[y = 1]$ 
324
325 '''
326 next
327 '''
328      $y^2 + (-8)y + 12 = 0$ 
329
330     #need two numbers that multiply to  $\rightarrow 12$  and add up to  $\rightarrow (-8)$ 
331      $a + b = (-8)$ 
332      $a * b = 12$ 
333
334      $(y + (-6))(y + (-2)) = 0$ 
335      $y + (-6) = 0$  or  $y + (-2) = 0$ 
336      $y = 6$  or  $y = 2$ 
337      $[y = 6 \text{ and } 2]$ 
338
339 '''
340 next
341 '''
342      $9 + 3x \geq (-6)$ 
343      $(-9) \quad (-9)$ 
344     #-----
345
346      $3x \geq (-15)$ 
347      $/ 3 \quad / 3$ 
348     #-----

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349
350      $[x \geq (-5)]$ 
351
352 '''
353 next
354 '''
355      $(x / 6) + (-2) < (-8)$ 
356          $+ 2 \quad + 2$ 
357     #-----
358
359      $(x / 6) < (-6)$ 
360          $* 6 \quad * 6$ 
361     #-----
362
363      $[x < (-36)]$ 
364
365 '''
366 next
367 '''
368      $9 > (-3) + 4x$ 
369      $+ 3 \quad + 3$ 
370     #-----
371
372      $12 > 4x$ 
373      $/ 4 \quad / 4$ 
374     #-----
375
376      $[3 > x]$ 
377
378 '''
379 next
380 '''
381     # $\infty$  always has a (parenthesis), never a [bracket]
382
383      $x \leq (-4) =$ 
384      $()-\infty, -4[$ 
385
386     #' $-\infty$ ' because it's all the negative numbers up to and including  $(-4)$ 
387
388 '''
389 next
390 '''
391      $x \geq (-5) =$ 
392      $[]-5, \infty()$ 
393
394 '''
395 next
396 '''
397      $(-10) + (-3v) < 8$ 
398      $+ 10 \quad + 10$ 

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399 #-----
400
401  $(-3v) < 18$ 
402  $/ (-3) \quad / (-3)$ 
403 #-----
404
405  $[v > (-6)]$ 
406
407 #the inequality sign gets flipped when you → divide by a negative number
408
409 '''
410 next
411 '''
412  $(-2u) + 25 > 15$ 
413  $(-25) \quad (-25)$ 
414 #-----
415
416  $(-2u) > (-10)$ 
417  $/ (-2) \quad / (-2)$ 
418 #-----
419
420  $[u < 5]$ 
421
422 #the inequality sign gets flipped when you → divide by a negative number
423
424 '''
425 next
426 '''
427  $6 \leq 3w + (-12)$ 
428  $+ 12 \quad + 12$ 
429 #-----
430
431  $18 \leq 3w$ 
432  $/ 3 \quad / 3$ 
433 #-----
434
435  $[6 \leq w]$ 
436
437 '''
438 next
439 '''
440  $M = \{(-2), 1, 5, 6, 7\}$ 
441  $I = \{0, 1, 3, 7\}$ 
442
443 #union combines all elements from both sets, without duplicates
444  $M \cup I = \{(-2), 0, 1, 3, 5, 6, 7\}$ 
445
446 #intersection includes only elements that are in both sets
447  $M \cap I = \{1, 7\}$ 
448

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449 #the phrasing is (elements) not (terms) because → sets contain distinct elements, not
algebraic terms
450
451 '''
452 next
453 '''
454  $E = \{a, g, j\}$ 
455  $G = \{c, g, k\}$ 
456
457 #union combines all elements from both sets, without duplicates
458  $E \cup G = \{a, c, g, j, k\}$ 
459
460 #intersection includes only elements that are in both sets
461  $E \cap G = \{g\}$ 
462
463 '''
464 next
465 '''
466  $A = \{g, h, k\}$ 
467  $L = \{a, d, h\}$ 
468
469 #union combines all elements from both sets, without duplicates
470  $A \cup L = \{a, d, g, h, k\}$ 
471
472 #intersection includes only elements that are in both sets
473  $A \cap L = \{h\}$ 
474
475 '''
476 next
477 '''
478  $I = \{(-1), 2, 3, 4, 7, 8\}$ 
479  $H = \{2, 3, 5, 7\}$ 
480
481 #union combines all elements from both sets, without duplicates
482  $I \cup H = \{(-1), 2, 3, 4, 5, 7, 8\}$ 
483
484 #intersection includes only elements that are in both sets
485  $I \cap H = \{2, 3, 7\}$ 
486
487 '''
488 next
489 '''
490 #case-1 → the expressions are equal →
491
492  $| 4x + (-5) | = | 4x + (-9) |$ 
493
494  $4x + (-5) = 4x + (-9)$ 
495  $(-4x) \quad (-4x)$ 
496 #-----
497

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498     (-5) = (-9)
499     [no solution]
500
501 #case-2 → the expressions are opposites →
502 #we strip away the absolute value bars and force one side to be negative by multiplying by
503 (-1) →
504
505 | 4x + (-5) | = | 4x + (-9) |
506 | 4x + (-5) | = (-1) | 4x + (-9) |
507 4x + (-5) = (-1)(4x + (-9))
508
509 4x + (-5) = (-4x + 9)
510 + 4x          + 4x
511 -----
512 8x + (-5) = 9
513   + 5     + 5
514 -----
515
516 8x = 14
517 / 8   / 8
518 -----
519
520 x = 14 / 8
521 #divide both numerator and denominator by 2 →
522 [x = 7 / 4]
523
524 #the cases are like an 'OR' logic path → does case-1 have a solution? no. does case-2 have a
525 solution? yes → that solution becomes the only solution for the entire equation
526
527 '''
528 next
529 '''
530 #case-1 → the expressions are equal →
531
532 | 4x + (-7) | = | (-1) + (-5x) |
533
534 4x + (-7) = (-1) + (-5x)
535 + 5x          + 5x
536 -----
537 9x + (-7) = (-1)
538   + 7     + 7
539 -----
540
541 9x = 6
542 / 9   / 9
543 -----
544
545 x = 6 / 9

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546 #divide both numerator and denominator by 3 →
547 [x = 2 / 3]
548
549 #case-2 → the expressions are opposites →
550 #remove the absolute value bars and multiply one side by (-1) →
551
552 | 4x + (-7) | = | (-1) + (-5x) |
553 | 4x + (-7) | = (-1) | (-1) + (-5x) |
554
555 4x + (-7) = (-1)((-1) + (-5x))
556
557 4x + (-7) = 1 + 5x
558 (-5x)          (-5x)
559 -----
560
561 (-x) + (-7) = 1
562   + 7     + 7
563 -----
564
565 (-x) = 8
566 /(-1) /(-1)
567 -----
568
569 [x = (-8)]
570
571 #two solutions →
572 [x = 2 / 3]
573 [x = (-8)]
574
575 '''
576 next
577 '''
578 #case-1 → the expressions are equal →
579
580 | 3u + 6 | = | (-3) + (-3u) |
581
582 3u + 6 = (-3) + (-3u)
583 + 3u (-6) (-6) + 3u
584 -----
585
586 6u = (-9)
587 / 6   / 6
588 -----
589
590 u = (-9) / 6
591 #divide both numerator and denominator by 3 →
592 [u = (-3) / 2 ]
593
594 #case-2 → the expressions are opposites →
595 #remove the absolute value bars and multiply one side by (-1) →

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596
597 | 3u + 6 | = | (-3) + (-3u) |
598 | 3u + 6 | = (-1) | (-3) + (-3u) |
599
600 3u + 6 = (-1)((-3) + (-3u))
601
602 3u + 6 = 3 + 3u
603 (-3u) (-3u)
604 #-----
605
606 6 = 3
607 [no solution]
608
609 #one solution →
610 [u = (-3) / 2 ]
611
612 '''
613 next
614 '''
615 (-3) < 3x + 3 < 9
616 (-3) (-3) (-3)
617 #-----
618
619 (-6) < 3x < 6
620 / 3 / 3 / 3
621 #-----
622
623 (-2) < x < 2
624
625 '''
626 next
627 '''
628 4x + (-4) > 12 or 2x + 4 ≤ (-2)
629
630 4x + (-4) > 12
631 + 4 + 4
632 #-----
633
634 4x > 16
635 / 4 / 4
636 #-----
637 [x > 4]
638
639 or
640
641 2x + 4 ≤ (-2)
642 (-4) (-4)
643 #-----
644
645 2x ≤ (-6)

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```

646 / 2 / 2
647 #-----
648
649 [x ≤ (-3)]
650
651 #solution set →
652 [x ≤ (-3)] or [x > 4]
653
654 '''
655 next
656 '''
657 (-2) < 2x + (-2) ≤ 10
658 + 2 + 2 + 2
659 #-----
660
661 0 < 2x ≤ 12
662 / 2 / 2 / 2
663 #-----
664
665 [0 < x ≤ 6]
666
667 '''
668 next
669 '''
670 | y + (-3) | > 5
671 '''
672 greater than → great-OR than → [or] statement
673 '''
674 #positive direction →
675 y + (-3) > 5
676 + 3 + 3
677 #-----
678 [y > 8]
679
680 #negative direction →
681 y + (-3) < (-5)
682 + 3 + 3
683 #-----
684
685 [y < (-2)]
686
687 #solution set →
688 [y < (-2)] or [y > 8]
689
690 '''
691 next
692 '''
693
694 less than → less-th-AND → [and] statement
695 '''

```

```

696 | y + (-8) | ≤ 2
697
698 #positive direction →
699 y + (-8) ≤ 2
700 + 8 + 8
701 #-----
702 [y ≤ 10]
703
704 #negative direction →
705 y + (-8) ≥ (-2)
706 + 8 + 8
707 #-----
708
709 [y ≥ 6]
710
711 #solution set →
712 [y ≥ 6] and [y ≤ 10]
713 [6 ≤ y ≤ 10]
714
715 '''
716 next
717 '''
718 '''
719 less than → less-th-AND → [and] statement
720 '''
721 | w + 2 | < 4
722
723 #positive direction →
724 w + 2 < 4
725 (-2) (-2)
726 #-----
727 [w < 2]
728
729 #negative direction →
730 w + 2 > (-4)
731 (-2) (-2)
732 #-----
733 [w > (-6)]
734
735 #solution set →
736 [w > (-6)] and [w < 2]
737 [(-6) < w < 2]
738
739
740
741

```